

MOJTABA HOUBALLAH

Numerical Modeling Scientist | PhD in Informatics

Paris region, France | mojtaba.houballah@gmail.com

[Portfolio website](#)

PROFILE

Numerical modeling scientist with a PhD in Informatics and a background in optimization. Develops, calibrates, and evaluates process-based environmental models driven by climate and meteorological inputs. Experienced in scientific computing, large simulation workflows, site-based validation, and uncertainty analysis. Interested in applying these methods to renewable-energy resource and yield assessment.

TECHNICAL EXPERTISE

Numerical models: Mathematical formulation, process-based simulation, implementation, and validation.

Meteorological and climate data: Forcing preparation, quality control, large datasets, and site observations.

Calibration and uncertainty: Optimization, sensitivity analysis, data assimilation, and error propagation.

Scientific computing: Python, Fortran, R, shell scripting, Linux, and high-performance computing.

Collaboration and supervision: Interdisciplinary research, technical presentations, and student mentoring.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Sep 2023 - Aug 2026

LSCE / IPSL, France

- Developed and evaluated ORCHIDEE-CROP, integrating STICS formulations into a coupled land-surface model with carbon, nitrogen, water, and energy processes.
- Developed Fortran, Python, R, and shell workflows for meteorological and climate-data preparation, quality control, model execution, and analysis of large numerical outputs on HPC systems.
- Implemented a coupled nitrogen-water stress scheme; diagnosed simulation behavior across European sites with contrasting climate and management conditions.
- Applied ORCHIDAS to parameter optimization, sensitivity analysis, and assessment of parameter transferability; evaluated simulations against observations.
- Supervised a student project on maize parameter optimization with ORCHIDAS in 2026.

Postdoctoral Researcher

Oct 2021 - Jul 2023

ESE, Université Paris-Saclay, France

- Developed and calibrated AMG-Forest, a process-based soil-carbon model evaluated against observations from the French RENECOFOR forest-monitoring network.
- Analyzed parameter sensitivity and model behavior using soil-carbon observations and Rock-Eval-derived indicators.
- Compared performance across sites and contributed to a framework for coupling with CASTANEA.

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher

Jun 2020 - Jul 2021

INRAE-SILVA, Nancy, France

- Combined heterogeneous institutional datasets into regional resource-flow models using data reconciliation.
- Assessed measurement uncertainty and error propagation in regional biomass and carbon estimates.
- Compared resource-use scenarios with carbon-accounting tools in collaboration with researchers and regional stakeholders.

Doctoral Researcher

Dec 2016 - Dec 2019

Université Clermont Auvergne / INRAE, France

- Developed dynamic models of environmental and management systems, linking physical processes, decisions, and economic constraints.
- Applied optimization, control theory, and viability theory to identify management strategies that remain feasible under changing constraints.
- Designed long-term simulations and multicriteria assessments, applied to forest management; produced three first-author peer-reviewed publications.

EDUCATION

PhD in Informatics, 2019: Université Clermont Auvergne, France. Mathematical modeling of environmental systems using viability theory.

Master II in Optimization, 2016: Université Paris-Saclay, France.

Master I in Mathematics, 2015: Lebanese University, Beirut.

BSc in Pure Mathematics, 2014: Lebanese University, Nabatieh.

TECHNICAL COMMUNICATION AND MENTORSHIP

- Supervised Ernest Djmodi on crop parameter optimization using ORCHIDAS (2026) and Jean Weber on territorial wood-flow quantification (2021).
- Presented ORCHIDEE-CROP development at the STICS Conference, Belgium (2026), and forest soil-carbon modeling at the EGU General Assembly, Austria (2023).

SELECTED RESEARCH OUTPUTS

- Houballah, M. et al. Simulation of decadal variations in soil organic carbon stocks in French forests. *Geoderma*, 2026. [Published article](#).
- Houballah, M., Cordonnier, T., Mathias, J.-D. Maintaining or building roads? An adaptive management approach for preserving forest multifunctionality. *Forest Ecology and Management*, 537, 120957, 2023.
- Houballah, M. et al. Development of ORCHIDEE-CROP v1.0: a nitrogen-explicit agro-land-surface model and its evaluation across European wheat sites. Manuscript in preparation.

MODELING FRAMEWORKS

ORCHIDEE-CROP, ORCHIDEE, STICS, ORCHIDAS, AMG-Forest, AMG, Century, CASTANEA, Carbon Accounting Tool.